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# Module 3: Introduction to MongoDB


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# Introduction to MongoDB: What is MongoDB, Why MongoDB, Terms used in RDBMS and MongoDB, Data Types in MongoDB, MongoDB Query Language.

## 3.1 What is MongoDB?

MongoDB is

1. Cross-platform.

2. Open source.

3. Non-relational.

4. Distributed.

5. NoSQL.

6. Document-oriented data store.



## 3.2 WHY MongoDB?

Few of the major challenges with traditional RDBMS are dealing with large volumes of data, rich variety of data-particularly unstructured data, and meeting up to the scale needs of enterprise data.

The need is for a database that can scale out or scale horizontally to meet the scale requirements, has flexibility with respect to schema, is fault tolerant, is consistent and partition tolerant, and can be easily distributed over a multitude of nodes in a cluster.

1. Full index support

2. Rich query language

3. Auto sharding

4. Document oriented

5. High performance

6. Fast in-place updates

7. Replication

8. Easy scalability

9. High availability

3.2.1 Using Java Script Object Notation(JSON)

• JSON is extremely expressive.

• MongoDB actually does not use JSON but BSON – it is Binary JSON. It is an open standard. It is used to store complex data structures.

Let us trace the journey from .csv to XML to JSON:

Let us look at how data is stored in .csv file.

Assume that this data is about the employees of an organization named "XYZ".

As we can see below, the column values are separated using commas and the rows are separated by a carriage return.

John, Mathews, +123 4567 8900

Andrews, Symmonds, +4567890 1234

Mable, Mathews, +789 1234 5678

However it can be made slightly more legible by adding column heading.

FirstName, LastName, ContactNo

John, Mathews, +1234567 8900

Andrews, Symmonds, +4567890 1234

Mable. Mathews, +789 12345678

**Challenges with CSV Format**

1. Flat Structure: CSV works best with flat, non-repeating data.

2. Multiple Values Problem:

Some employees have multiple Office and Home contact numbers.

Some have multiple email addresses (2, 3, or more).

3. Merge Complexity:

When different departments use different CSV formats, merging becomes tedious and error-prone.

Field inconsistency and missing/extra columns further complicate consolidation.

**Why XML is Not Ideal for Simpler Use Cases**

Pros:

• XML supports complex and hierarchical data structures.

• Suitable for highly structured data formats.

Cons:

• Too verbose and heavy for simple data exchange.

• Requires definition of data structure using schemas or DTDs.

• Overkill for lightweight, frequently changing employee records.

**JSON as an Effective Alternative**

1. Extensible and lightweight.

2. Handles arrays/lists of data naturally (e.g., multiple contacts or emails).

3. Easy to read and write.

4. Excellent for web applications and APIs


```
{
        "FirstName": "John",
        "LastName": "Mathews",
        "ContactNo": ["+123 45678900", "+123 4444 5555"],
        "Emails": ["john@example.com", "john.mathews@work.com"]
},
{
        "FirstName": "Andrews",
        "LastName": "Symmonds",
        "ContactNo": ["+456 7890 1234", "+456 6666 7777"]
},
{
        "FirstName": "Mable",
```

```
        "LastName": "Mathews",

        "ContactNo": ["+789 1234 5678"]

}
```

JSON is very expressive. It provides the much needed ease to store and retrieve documents in their real form. The binary form of JSON is BSON. BSON is an open standard. In most cases it consumes less space as compared to the textbased JSON. There is yet another advantage with BSON. It is much easier and quicker to convert BSON to a programming language's native data format.

There are MongoDB drivers available for a number of programming languages such as C, C++, Ruby, PHP, Python, C#, etc., and each works slightly differently. Using the basic binary format enables the native data structures to be built quickly for each language without going through the hassle of first processing JSON.

### 3.2.2 Creating or generating a Unique key

• Each JSON document should have a unique identifier.

• It is the _id key.

• It is similar to the primary key in relational databases.

• This facilitates search for documents based on the unique identifier.

• An index is automatically built on the unique identifier.

• It is your choice to either provide unique values yourself or have the mongo shell generate the same.

#### 3.2.2.1 Database

It is a collection of collections. In other words, it is like a container for collections. It gets created the first time that your collection makes a reference to it. This can also be created on demand. Each database gets its own set of files on the file system. A single MongoDB server can house several databases.

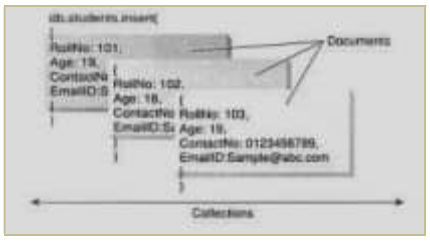
### 3.2.2.2 Collection

A collection is analogous to a table of RDBMS. A collection is created on demand. It gets created the first time that you attempt to save a document that references it. A collection exists within a single database. A collection holds several MongoDB documents. A collection does not enforce a schema. This implies that documents within a collection can have different fields. Even if the documents within a collection have same fields, the order of the fields can be different.

### 3.2.2.3 Document

A document is analogous to a row/record/tuple in an RDBMS table. A document has a dynamic schema. This implies that a document in a collection need not necessarily have the same set of fields/key-value pairs.

Shown in Figure below is a collection by the name "students" containing three documents.




### 3.2.3 Support for Dynamic Queries

MongoDB has extensive support for dynamic queries. This is in keeping with traditional RDBMS wherein we have static data and dynamic queries.

CouchDB, another document-oriented, schema-less NoSQL data-base and MongoDB's biggest competitor, works on quite the reverse philosophy. It has support for dynamic data and static queries.

**MongoDB: Dynamic Queries, Static Data**

MongoDB allows you to build dynamic queries using a rich and expressive query language.

You can query on any field, including nested documents and arrays.

This aligns with the traditional RDBMS approach, where the data structure (schema) is fixed, and the queries are dynamic.

Example: You can construct queries at runtime, filter based on various fields, and use advanced operators ($gt, $in, $or, etc.).

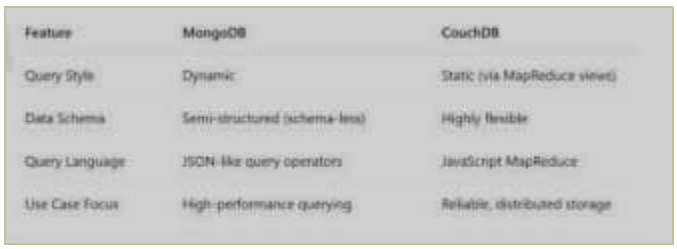
**CouchDB: Static Queries, Dynamic Data**

CouchDB uses MapReduce views to query data.

Once a view (which is essentially a query) is defined, it becomes static—you cannot alter it dynamically without redefining the view.

The data is more flexible, and each document can have a completely different structure.

This allows more flexibility in storing heterogeneous data but limits ad-hoc querying unless pre-defined.



| Feature | MongoDB | CouchDB |
|---|---|---|
| Query Style | Dynamic | Static (via MapReduce views) |
| Data Schema | Semi-structured (schema-less) | Highly flexible |
| Query Language | JSON-like query operators | JavaScript MapReduce |
| Use Case Focus | High-performance querying | Reliable, distributed storage |

### 3.2.4 Storing Binary Data

MongoDB provides GridFS to support the storage of binary data. It can store up to 4 MB of data. This usually suffices for photographs (such as a profile picture) or small audio clips. However, if one wishes to store movie clips,

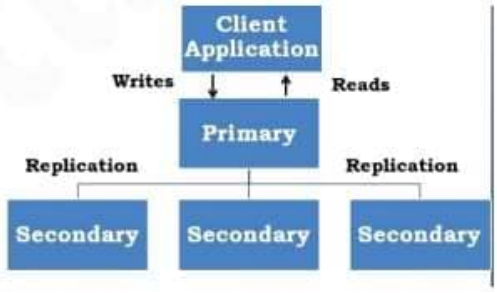
**MongoDB has another solution.**

It stores the metadata (data about data along with the context information) in a collection called "file". It then breaks the data into small pieces called chunks and stores it in the "chunks" collection. This process takes care about the need for easy scalability.

### 3.2.5 Replication

**Why replication?**

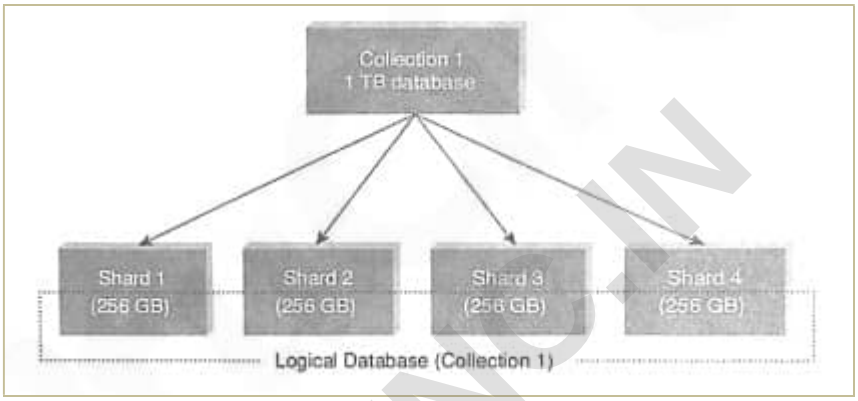
It provides data redundancy and high availability. It helps to recover from hardware failure and service interruptions. In MongoDB, the replica set has a single primary and several secondaries. Each write request from the client is directed to the primary. The primary logs all write requests into its Oplog (operations log). The Oplog is then used by the secondary replica members to synchronize their data. This way there is strict adherence to consistency Refer Figure. The clients usually read from the primary. However, the client can also specify a read preference that will then direct the read operations to the secondary.




### 3.2.6 Sharding

Sharding is akin to horizontal scaling. It means that the large dataset is divided and distributed over multiple servers or shards. Each shard is an independent database and collectively they would constitute a logical database. The prime advantages of sharding are as follows: 1. Sharding reduces the amount of data that each shard needs to store and

manage. For example, if the dataset was 1 TB in size and we were to distribute this over four shards, each shard would house just 256 GB data. As the cluster grows, the amount of data that each shard will store and manage will decrease. 2. Sharding reduces the number of operations that each shard handles. For example, if we were to insert data, the application needs to access only that shard which houses that data.




### 3.2.7 Updating Information In-Place

MongoDB updates the information in-place. This implies that it updates the data wherever it is available. It does not allocate separate space and the indexes remain unaltered.

MongoDB is all for lazy-writes. It writes to the disk once every second. Reading and writing to disk is a slow operation as compared to reading and writing from memory. The fewer the reads and writes that we perform to the disk, the better is the performance. This makes MongoDB faster than its other competitors who write almost immediately to the disk. However, there is a tradeoff. MongoDB makes no guarantee that data will be stored safely on the disk.

## 3.3 Terms used in RDBMS and MongoDB

### Structural Differences

Here is the tabular format of the image you uploaded:

| RDBMS Terms | MongoDB Terms | Description |
|---|---|---|

| RDBMS Terms | MongoDB Terms | Description |
|---|---|---|
| Database | Database | A logical grouping of data. Both RDBMS and MongoDB use this term. |
| Table | Collection | A group of related records. In MongoDB, a collection stores multiple documents instead of rows. |
| Row | Document | In MongoDB, a document is a JSON-like object (BSON) that holds data in key-value pairs. |
| Column | Field | A specific attribute of data in RDBMS is called a field in MongoDB. |
| Primary Key | _id Field | Every MongoDB document has a unique _id field, which acts as the primary key. |

## Relationships and Joins

| RDBMS Terms | MongoDB Equivalent | Description |
|---|---|---|
| Foreign Key | Reference (Manual or `$lookup`) | In MongoDB, relationships can be represented by storing references to documents in other collections. |
| JOIN Operation | Embedding or `$lookup` | RDBMS uses JOIN to fetch related data, while MongoDB supports embedding related documents or using `$lookup` for similar functionality. |



## Indexing and Querying

| RDBMS Terms | MongoDB Equivalent | Description |
|---|---|---|
| Index | Index | Both RDBMS and MongoDB use indexes to speed up queries. MongoDB supports single-field, compound, text, and geospatial indexes. |
| SQL (SELECT, WHERE, etc.) | Find Query (db.collection.find()) | MongoDB uses a JavaScript-like query syntax instead of SQL. |
| Aggregate Functions (SUM, AVG, COUNT etc.) | Aggregation Pipeline | MongoDB's aggregation framework replaces SQL functions with operations like $sum, $avg, etc., processed through a pipeline. $group, $sum, $avg, etc. |

### Transactions and Consistency

| RDBMS Terms | MongoDB Equivalent | Description |
|---|---|---|
| ACID Transactions | Multi-document Transactions | RDBMS supports ACID compliance by default, while MongoDB introduced multi-document transactions in version 4.0 for similar behavior. |
| Commit & Rollback | Session-Based Transactions | In MongoDB, transactions work using session-based operations. |

Session-based operation refers to processes where a user's interactions are tracked across multiple requests using a unique session ID. It allows continuity, like maintaining login state or a shopping cart, throughout a user's visit.

### Performance and Scaling

| RDBMS Terms | MongoDB Equivalent | Description |
|---|---|---|
| Vertical Scaling | Horizontal Scaling | RDBMS typically scales vertically by upgrading hardware, whereas MongoDB scales horizontally using sharding. |
| Partitioning | Sharding | servers in MongoDB is handled by sharding |
| Replication (Master-Slave) | Replication (Replica Sets) | Both databases support replication, but MongoDB uses Replica Sets for automatic failover and redundancy. |
| | | |

### Backup and Recovery

| RDBMS Terms | MongoDB Equivalent | Description |
|---|---|---|
| Backup & Restore (mysqldump, pg_dump) | mongodump & mongorestore | MongoDB provides tools like mongodump and mongorestore for backups. |
| Point-in-Time Recovery | Oplog Replay | In MongoDB, oplog (operations log) enables point-in-time recovery for replica sets |

| | | |
|---|---|---|

### Data Integrity and Constraints

| RDBMS Terms | MongoDB Equivalent | Description |
|---|---|---|
| Constraints (NOT NULL, UNIQUE, CHECK, FOREIGN KEY) | Schema Validation & Document Validation | MongoDB allows schema validation rules but is not as strict as RDBMS constraints. |
| Triggers | Change Streams | MongoDB supports change streams, which notify applications of data changes in real time. |

### Security and Authentication

| RDBMS Terms | MongoDB Equivalent | Description |
|---|---|---|
| User Roles & Permissions | Role-Based Access Control (RBAC) | MongoDB uses RBAC for fine-grained security. |
| Authentication (LDAP, Kerberos, etc.) | Authentication with LDAP, Kerberos, SCRAM | MongoDB supports authentication methods like LDAP, Kerberos, and SCRAM-SHA. |
| Encryption | Encryption at Rest & In Transit | MongoDB supports TLS/SSL for encrypted connections and AES-256 for data encryption at rest. |



## 3.3.1 Create Database

### *Creating a Database*

Syntax: use DATABASE_Name

Example: To create a database named myDB,

use: use myDB

Output: switched to db myDB

### *Verifying the Current Database*

To check which database you are currently using:

db

Output: myDB

### *Listing All Databases*

To list all existing databases:

show dbs

Output (example):

admin (empty)

local 0.078GB

test 0.078GB

•The newly created database (e.g., myDB) does not appear in the list from show dbs until it contains at least one document.

• The default database in MongoDB is test.

If no database is explicitly created, any collections inserted will be stored in the test database.

## 3.3.2 Drop Database

Syntax to Drop a Database

db.dropDatabase();

Steps to Drop a Specific Database (e.g., "myDB")

1. First, switch to the database you want to drop:

2. use myDB;

3. Then execute the drop command:

4. db.dropDatabase();

Confirmation Message

After running the command, MongoDB returns:

{ "dropped" : "myDB", "ok" : 1 }

• Always ensure you are in the correct database before executing db.dropDatabase().

• If no database is selected, MongoDB will drop the default database test.

**Data Types in MongoDB**

| | |
|---|---|
| String | Must be UTF-8 valid.<br>Most commonly used data type. |
| Integer | Can be 32-bit or 64-bit (depends on the server). |
| Boolean | To store a true/false value. |
| Double | To store floating point (real values). |
| Min/Max keys | To compare a value against the lowest or highest BSON elements. |
| Arrays | To store arrays or list or multiple values into one key. |
| Timestamp | To record when a document has been modified or added. |
| Null | To store a NULL value. A NULL is a missing or unknown value. |
| Date | To store the current date or time in Unix time format. One can create object of date and pass day, month and year to it. |
| Object ID | To store the document's id. |
| Binary data | To store binary data (images, binaries, etc.). |
| Code | To store javascript code into the document. |
| Regular expression | To store regular expression. |

**1. String**

In MongoDB, strings must be UTF-8 encoded.

Ex: { "name": "John Doe" }

**2. Integer**

MongoDB differentiate between 32-bit(Int32) and 64-bit(Int64) integer

Ex: { "age": 25 }

**3. Double**

Ex: {“Salary” : 1900.54}

**4. Boolean**

Ex: {“isActive : true”}

**5. Array**

Stores multiple values in a single field

Ex: { "skills": ["JavaScript", "Python", "MongoDB"] }

**6. Object(Embedded document or subdocument )**

Stores a document inside another document.

Ex : { "name": "Alice", "address": { "city": "New York", "zip": "10001" }}

**7. ObjectId**

A 12-byte identifier (timestamp, machine ID, process ID, and counter).

Ex: { "_id": ObjectId("507f1f77bcf86cd799439011") }

**8. Date**

Default format: ISODate("YYYY-MM-DDTHH:MM:SSZ").

Ex: { "createdAt": ISODate("2024-03-25T10:30:00Z") }

**9. Null**

Ex: { "deletedAt": null }

**10. Binary Data**

Stores binary data such as images, audio, or encrypted data.

Ex : { "profilePicture": BinData(0, "base64EncodedData") }

**11. Regular Expression**

Stores and queries strings using regex patterns.

Ex: { "pattern": /mongodb/i }

**12. JavaScript Code**

**E**x: { "script": function() { return "Hello MongoDB"; } }

**13. JavaScript with Scope**

Similar to javascript, but allows defining scope (variables) for the script.

Ex: { "script": { "$code": "function(x) { return x * 2; }", "$scope": { "x":

10 } }}

**14. Timestamp**

Stores a high-precision timestamp (used for internal MongoDB

operations).

Ex: { "createdAt": Timestamp(1618928492, 1) }

**15. Decimal128**

High-precision 128-bit decimal numbers (useful for financial calculations).

Ex: { "amount": NumberDecimal("1234.5678") }

**16. MinKey & MaxKey**

MinKey: Represents the lowest possible value in MongoDB (useful for sorting).

MaxKey: Represents the highest possible value.

Ex : { "lowestValue": MinKey(), "highestValue": MaxKey() }

**Useful MongoDB Shell Commands**

**1. To report the name of the current database:**

db

Example Output:

test

**2. To display the list of all databases:**

show dbs

Example Output:

admin (empty)

local 0.078GB

myDB1 0.078GB

**3.To switch to a new database (e.g., myDB1):**

use myDB1

Output:

switched to db myDB1

**4.To display the list of collections (tables) in the current database:**

show collections

Example Output:

system.indexes

system.js

**5.To display the current version of the MongoDB server:**

db.version()

Example Output:

### 2.6.1 Consider a table “Students” with the following columns:

1. **StudRoll No**
2. **StudName**
3. **Grade**
4. **Hobbies**
5. **DOJ**

**Before we get into the details of CRUD operations in MongoDB, let us look at how the statements are written in RDBMS and MongoDB.**

*Useful MongoDB Shell Commands*

**To report the name of the current database:**

**db**

**Example Output:**

**test**

**To display the list of all databases:**

**show dbs**

**Example Output:**

**admin (empty)**

**local 0.078GB**

**myDB1 0.078GB**

**To switch to a new database (e.g., myDB1):**

**use myDB1**

**Output:**

**switched to db myDB1**

**To display the list of collections (tables) in the current database:**

**show collections**

**Example Output:**

**system.indexes**

**system.js**

**To display the current version of the MongoDB server:**

**db.version()**

**Example Output:**

### 2.6.1 Consider a table "Students" with the following columns:

1. StudRoll No
2. StudName
3. Grade
4. Hobbies 5. DOJ

**Before we get into the details of CRUD operations in MongoDB, let us look at**

**how the statements are written in RDBMS and MongoDB.**

| | RDBMS | MongoDB |
|---|---|---|
| Insert | Insert into Students (StudRollNo, StudName, Grade, Hobbies, DOJ) Values ('S101', 'Simon David', 'VII', 'Net Surfing', '10-Oct-2012') | db.Students.insert({_id:1, StudRollNo: 'S101', StudName: 'Simon David', Grade: 'VII', Hobbies: 'Net Surfing', DOJ: '10-Oct-2012'}); |
| Update | Update Students set Hobbies ='Ice Hockey' where StudRollNo ='S101' | db.Students.update({StudRollNo: 'S101'}, {$set: {Hobbies : 'Ice Hockey'}}) |
| | Update Students Set Hobbies ='Ice Hockey' | db.Students.update({},{$set: {Hobbies: 'Ice Hockey' }}, {multi:true}) |
| Delete | Delete from Students where StudRollNo = 'S101' | db.Students.remove ({StudRollNo : 'S101'}) |
| | Delete From Students | db.Students.remove({}) |
| Select | Select * from Students | db.Students.find() db.Students.find().pretty() |
| | Select * from students where StudRollNo = 'S101' | db.Students.find({StudRollNo: 'S101'}) |
| | Select StudRollNo, StudName, Hobbies from Students | db.Students.find({},{StudRollNo:1, StudName:1, Hobbies:1, _id:0}) |
| | Select StudRollNo, StudName, Hobbies from Students where StudRollNo = 'S101' | db.Students.find({StudRollNo: 'S101'}, {StudRollNo : 1, StudName: 1, Hobbies : 1, _id:0}) |

| RDBMS | MongoDB |
|---|---|
| Select StudRollNo, StudName, Hobbies From Students Where Grade ='VII' and Hobbies ='Ice Hockey' | db.Students.find({Grade: 'VII' , Hobbies: 'Ice Hockey'}, {StudRollNo : 1, StudName: 1, Hobbies : 1, _id:0}) |
| Select StudRollNo, StudName, Hobbies From Students Where Grade ='VII' or Hobbies = 'Ice Hockey' | db.Students.find({ $or: [{Grade: 'VII' , Hobbies: 'Ice Hockey'}], StudRollNo : 1, StudName: 1, Hobbies : 1, _id:0}) |
| Select * From Students Where StudName like 'S%' | db.Students.find({ StudName: /^S/}).pretty() |

## 3.5 MongoDB Query Language

### CRUD (Create, Read Update, and Delete) operations in MongoDB

*Create → Creation of data is done using insert() or update() or save() method.*

*Read → Reading the data is performed using the find() method.*

*Update → Update to data is accomplished using the update() method with*

*UPSERT set to false.*

*Delete → a document is Deleted using the remove() method.*

## Creating and Dropping Collections

### Creating a Collection

**Objective: Create a collection named "Person".**

**Step 1 – View existing collections:**

```
show collections
```

**Example output:**

```
Students
food
system.indexes
system.js
```

**Step 2 – Create the new collection:**

```
db.createCollection("Person")
```

**Output:**

```
{ "ok" : 1 }
```

**Outcome – View updated collections:**

**show collections**

**Example output after creation:**

**Person**

**Students**

**food**

**system.indexes**

**system.js**

## Dropping a Collection

**Objective: Drop the collection named "food".**

### Step 1 – Check current collections:

**show collections**

**Example output:**

**Person**

**Students**

**food**

**system.indexes**

**system.js**

**Step 2 – Drop the collection:**

**db.food.drop()**

**Output:**

**true**

**Outcome – View updated collections:**

**show collections**

**Example output after dropping:**

**Person**

**Students**

**system.indexes**

**system.js**



### 3.5.1 Insert Method

#### 1. Create Collection and Insert Document

**Objective: Create a collection named Students and insert a document.**

Check existing collections:

show collections

Insert first document:

db.Students.insert({_id:1, StudName:"Michelle Jacintha", Grade:"VII",

Hobbies:"Internet Surfing"})

Verify insertion:

db.Students.find().pretty()

### 2. Insert Another Document

**Insert second document:**

db.Students.insert({_id:2, StudName:"Mabel Mathews", Grade:"VII",

Hobbies:"Baseball"})

Verify with pretty():

db.Students.find().pretty()

### 3. Conditional Insert/Update with upsert

**Objective: Insert Aryan David only if not already in the collection. If present,**

update his hobbies.

Check current documents:

db.Students.find().pretty()

Insert using upsert:

db.Students.update({_id:3}, {StudName:"Aryan David", Grade:"VII",

$set:{Hobbies:"Skating"}}, {upsert:true})

Confirm insertion:

db.Students.find().pretty()

### 4. Update Existing Document

Objective: Update Aryan David's hobbies from "Skating" to "Chess".

Update using upsert (will update if exists, insert otherwise):

db.Students.update({_id:3}, {StudName:"Aryan David", Grade:"VII",

$set:{Hobbies:"Chess"}}, {upsert:true})

### 5. Insert Document Using save()

**Objective: Insert Vamsi Bapat without specifying _id.**

Save document:

db.Students.save({StudName:"Vamsi Bapat", Grade:"VII"})

Check final documents:

db.Students.find().pretty()

## 3.5.2 save() method

Inserts a new document if no document with the specified _id exists. If the document exists, it replaces the existing one.

**Objective:**

Insert the document of "Hersch Gibbs" into the Students collection using the

update() method with the upsert option.

**Step 1: Check existing documents in the "Students" collection**

Shows the existing documents with their _id, StudName, Grade, and Hobbies.

**Step 2: Use update with upsert: false**

```
db.Students.update(
        {_id:4, StudName:"Hersch Gibbs", Grade:"VII"},
        {$set: {Hobbies: "Graffiti"}},
        {upsert: false}
);
```

• No document is inserted because a document with _id:4 doesn't exist.

• Result shows nUpserted: 0 meaning no document was inserted.

**Step 3: Use update with upsert: true**

```
db.Students.update(
        {_id:4, StudName:"Hersch Gibbs", Grade:"VII"},
        {$set: {Hobbies: "Graffiti"}},
        {upsert: true}
);
```

• A new document with _id:4 is inserted.

• Result shows nUpserted: 1, meaning one document was inserted.

**Step 4: Confirm the new document**

db.Students.find() now shows the new document of "Hersch Gibbs" with the Hobbies: "Graffiti" included.

**3.5.3 Add a new field to an existing document-Update Method**

**Syntax of update method**

**db.students.update(**

{Age: {$gt: 18}}, // Update Criteria (which documents to update)

{$set: {Status: "A"}}, // Update Action (what to update/set)

{multi: true} // Update Option (update multiple documents)

**)**

**Objective:**

**Add a new field "Location" with value "Newark" to the document with _id:4 in the "Students" collection.**

Input:

Check the document with _id:4 before updating:

db.Students.find({_id:4}).pretty();

**Output:**

```
{
        "_id": 4,
        "Grade": "VII",
        "StudName":"Hersch Gibbs",
        "Hobbies": "Graffiti"
}
```

**Act:**

Add the new field "Location" with the value "Newark":

```
db.Students.update(
        {_id:4},
        {$set: {Location: "Newark"}}
);
```

**Output shows:**

```
{
        "nMatched": 1,
        "nUpserted": 0,
        "nModified": 1
}
```

Outcome:

Confirm the new field has been added:

db.Students.find({_id:4}).pretty();

**Output:**

{

"_id": 4,

"Grade": "VII",

"StudName": "Hersch Gibbs",

"Hobbies": "Graffiti",

"Location": "Newark"

}



### 3.5.4 Removing an Existing Field from an Existing Document – Remove Method

**Objective:**

To remove the field "Location" with the value "Newark" from a document with _id: 4 in the Students collection.

**Input:**

Inspect the current document:

db.Students.find({_id:4}).pretty()

**Output:**

{

"_id": 4,

"Grade": "VII",

"StudName": "Hersch Gibbs",

"Hobbies": "Graffiti",

"Location": "Newark"

}

**Act:**

Execute the update command to remove the "Location" field:

db.Students.update({_id:4}, { $unset: { Location: "Newark" } })

**This uses:**

• update to modify the document,

• $unset to remove the "Location" field (value is ignored, key matters).

**Output:**

WriteResult({ "nMatched" : 1, "nUpserted" : 0, "nModified" : 1 })

Outcome:

Verify the document again:

db.Students.find({_id:4}).pretty()

**Result:**

{

"_id": 4,

"Grade": "VII",

"StudName":"Hersch Gibbs",

"Hobbies": "Graffiti"

}

**The "Location" field has been successfully removed.**

**1. Removing a Document with remove() Method**

db.Students.remove({Age: {$gt: 18}})

• Removes all documents in the Students collection where the Age isgreater than 18.

**2. Removing a Field from a Document with $unset**

db.Students.update({_id: 4}, {$unset: {Location: "Newark"}})

• Removes the Location field from the document with _id: 4.

**3. Updating Documents with $set**

```
db.Students.update({_id: 4}, {$set: {Grade: "X"}})
```

• Updates the Grade field of the document with _id: 4 to "X".

**4. Adding New Fields**

```
db.Students.update({_id: 4}, {$set: {Sports: "Football"}})
```

• Adds a new field Sports with the value "Football" to the document with _id: 4.

**5. Using multi: true to Update Multiple Documents**

```
db.Students.update(
{Grade: "VII"},
{$set: {Sports: "Cricket"}},
{multi: true}
)
```

• Updates all documents where Grade is "VII" by setting Sports to "Cricket".

**6. Replacing an Entire Document**

```
db.Students.replaceOne(
{_id: 4},
{
_id: 4,
```

Grade:"X",

StudName: "Hersch Gibbs",

Hobbies: "Graffiti",

Sports: "Football"

}

)

• Replaces the entire document with _id: 4.

**7. Upsert Operation (update + upsert: true)**

db.Students.update(

{_id: 5},

{$set: {StudName: "Paul Adams", Grade: "VIII"}},

{upsert: true}

)

• If a document with _id: 5 doesn't exist, MongoDB inserts it with the specified fields.

### 3.5.7 Finding Elements Based on Some Criteria – findOne() Method

**Objective:**

To retrieve a single document from the "Students" collection where a specific

field (e.g., Grade) matches a value.

Command:

db.Students.findOne({Grade: "VII"})

• This will return only one document (not all matches).

• Equivalent in SQL:

SELECT * FROM Students WHERE Grade = 'VII' LIMIT 1;

Finding Specific Elements – find() with Projections

Objective:

Retrieve only selected fields from matching documents.

Command:

db.Students.find({Grade: "VII"}, {StudName: 1, _id: 0})

• This will return only the StudName of students in Grade "VII".

• _id: 0 hides the _id field.

• Equivalent in SQL:

SELECT StudName FROM Students WHERE Grade = 'VII';

**Sorting Results – sort() Method**

Objective:

Sort documents in ascending or descending order by a specified field.

Commands:

db.Students.find().sort({Grade: 1}) // Ascending

db.Students.find().sort({Grade: -1}) // Descending

• Equivalent in SQL:

SELECT * FROM Students ORDER BY Grade ASC;

SELECT * FROM Students ORDER BY Grade DESC;

Limiting Results – limit() Method

Objective:

Retrieve only a certain number of documents.

Command:

db.Students.find().limit(3)

• Returns the first 3 documents from the collection.

• Equivalent in SQL:

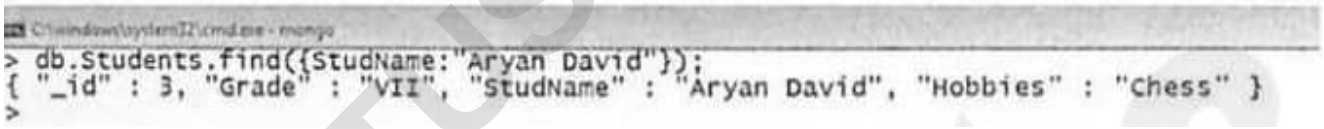
SELECT * FROM Students LIMIT 3;

### 3.5.5 Finding Documents based on Search Criteria - Find Method

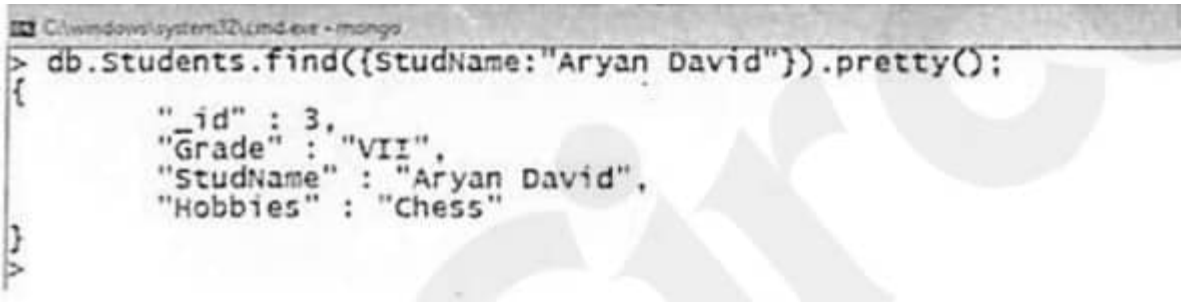
Objective: To search for documents from the "Students" collection based on certain search criteria. Input: Check the documents in the "Students" collection before proceeding.


**Act:**

**Find the document wherein the "StudName" has value "Aryan David". db.Students.find({StudName:"Aryan David"});**




**To format the above output, use the pretty() method: db.Students.find({StudName:"Aryan David"}).pretty();**




**RDBMS equivalent: Select * From Students Where StudName like 'Aryan David'; Objective: To display only the StudName from all the documents of the**

**Student's collection. The identifier "_id" should be suppressed and NOT displayed. Act: db.Students.find({}, {StudName: 1,_id:0});**

**Outcome:**


**RDBMS equivalent:**

Select StudName From Students;

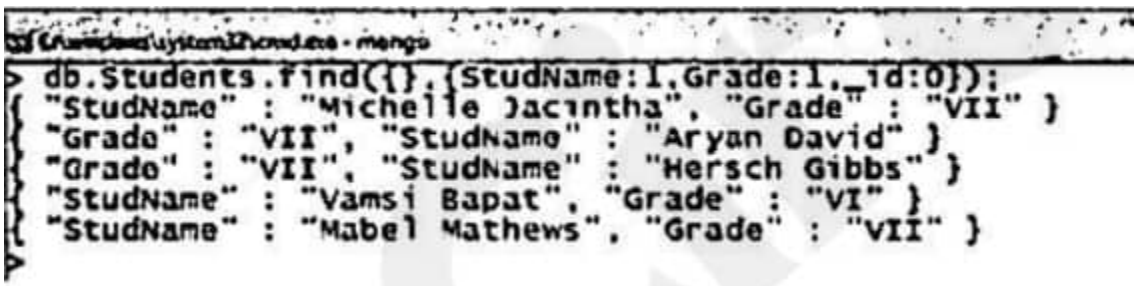
Objective:

To display only the StudName and Grade from all the documents of the Students collec- tion. The identifier _id should be suppressed and NOT displayed.

Act:

db.Students.find({}, {StudName:1,Grade: 1,_id:0});

**Outcome:**




**RDBMS equivalent:**

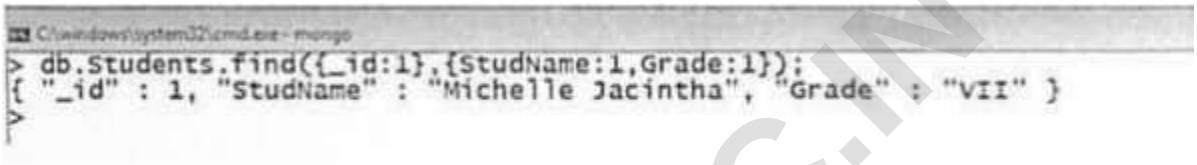
Select StudName, Grade From Students;

Objective: To display the StudName, Grade as well the identifier, id from the document of the Students collection where the _id column is 1.

Act:

```
db.Students.find({_id:1},{StudName:1,Grade:1});
```

**Outcome:**

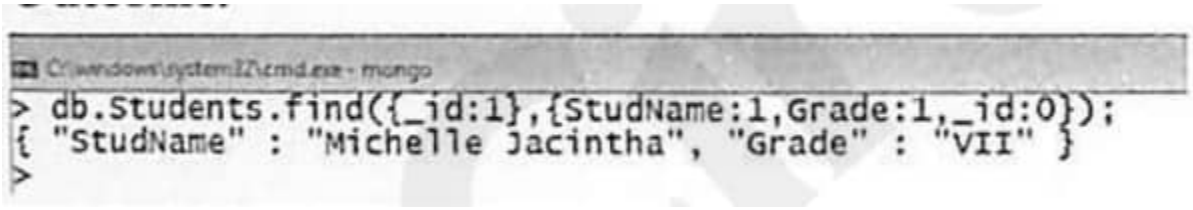



**RDBMS equivalent:**

Select StudRoll No, StudName, Grade From Students Where StudRollNo = '1';
Objective: To display the StudName and Grade from the document of the Students collection where the_id column is 1. The id field should NOT be displayed.

Act:

```
db.Students.find({_id:1}, {StudName:1,Grade:1,_id:0});
```




**RDBMS equivalent:**

Select StudName, Grade

From Students

Where StudRollNo like '1';

***Relational operators available to use in the search criteria:***

| | |
|---|---|
| $eq | → equal to |
| $ne | → not equal to |
| $gte | → greater than or equal to |
| $lte | → less than or equal to |
| $gt | → greater than |
| $lt | → less than |

**Objective: To find those documents where the Grade is set to 'VII'.**

**Act:**

**db.Students.find({Grade: {$eq:'VII'}}).pretty();**

```
> db.Students.find({Grade:{$eq:'VII'}}).pretty();
{
        "_id" : 1,
        "StudName" : "Michelle Jacintha",
        "Grade" : "VII",
        "Hobbies" : "Internet Surfing"
}
{
        "_id" : 3,
        "Grade" : "VII",
        "StudName" : "Aryan David",
        "Hobbies" : "Chess"
}
{
        "_id" : 4,
        "Grade" : "VII",
        "StudName" : "Hersch Gibbs",
        "Hobbies" : "Graffiti"
}
{
        "_id" : 2,
        "StudName" : "Mabel Mathews",
        "Grade" : "VII",
        "Hobbies" : "Baseball"
}
```

**RDBMS Equivalent:**

Select * From Students

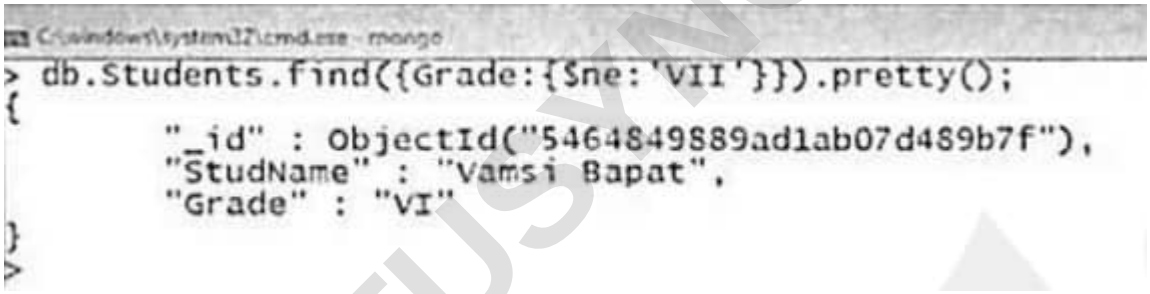
Where Grade like 'VII';

Objective:

To find those documents where the Grade is NOT set to 'VII'.

Act:

db.Students.find({Grade: {$ne: 'VII'}}).pretty();




**RDBMS Equivalent:**

Select *

From Students

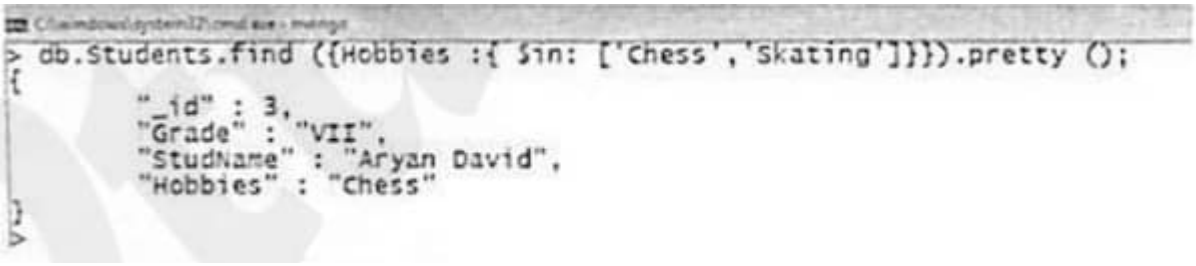
Where Grade <> 'VII';

Objective: To find those documents from the Students collection where the

Hobbies is set to either 'Chess' or is set to 'Skating'.

Act:

db.Students.find ({Hobbies :{ $in: ['Chess', 'Skating']}}).pretty ();




**RDBMS Equivalent:**

Select *

From Students

Where Hobbies in ('Chess', 'Skating');

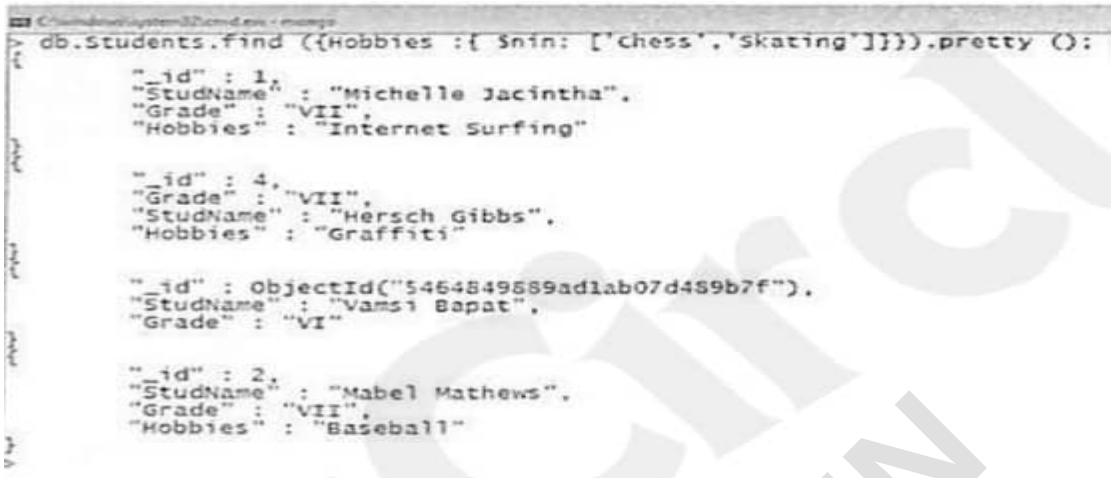
Objective: To find those documents from the Students collection where the

Hobbies is set neither to 'Chess' nor is set to 'Skating'.

Act:

db.Students.find({Hobbies :{ $nin: ['Chess','Skating']}}).pretty ();

**Outcome:**



```
> db.Students.find ({Hobbies :{ $nin: ['Chess','Skating']}}).pretty ():
{
        "_id" : 1,
        "StudName" : "Michelle Jacintha",
        "Grade" : "VII",
        "Hobbies" : "Internet Surfing"

        "_id" : 4,
        "Grade" : "VII",
        "StudName" : "Hersch Gibbs",
        "Hobbies" : "Graffiti"

        "_id" : ObjectId("5464849689ad1ab07d489b7f"),
        "StudName" : "Vamsi Bapat",
        "Grade" : "VI"

        "_id" : 2,
        "StudName" : "Mabel Mathews",
        "Grade" : "VII",
        "Hobbies" : "Baseball"
}
>
```

**RDBMS Equivalent:**

Select *

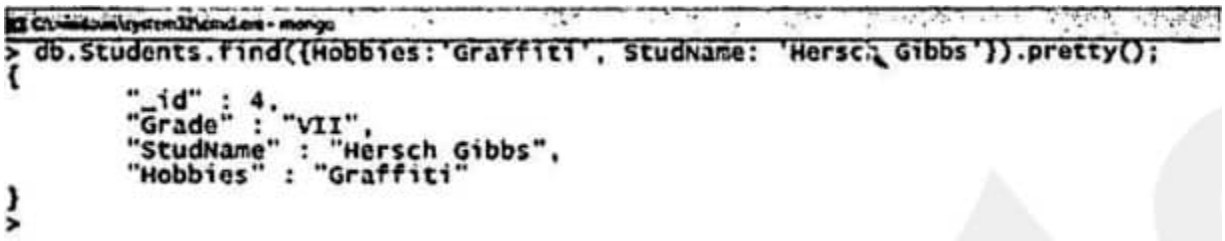
From Students

Where Hobbies not in ('Chess', 'Skating');

Objective: To find those documents from the Students collection where the Hobbies is set to 'Graffiti' and the StudName is set to 'Hersch Gibbs' (AND condition).

Act:

db.Students.find({Hobbies:'Graffiti', StudName: 'Hersch Gibbs'}).pretty();



```
> db.Students.find({Hobbies:'Graffiti', StudName: 'Hersch Gibbs'}).pretty();
{
        "_id" : 4,
        "Grade" : "VII",
        "StudName" : "Hersch Gibbs",
        "Hobbies" : "Graffiti"
}
>
```

**RDBMS Equivalent:**

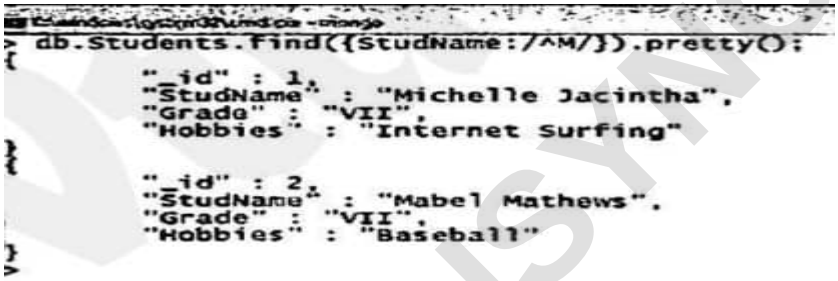
Select * From Students

Where Hobbies like 'Graffiti' and StudName like 'Hersch Gibbs';

Objective: To find documents from the Students collection where the StudName begins with "M".

Act:

db.Students.find({StudName:/^M/}).pretty(); Outcome:




**RDBMS Equivalent:**

Select *

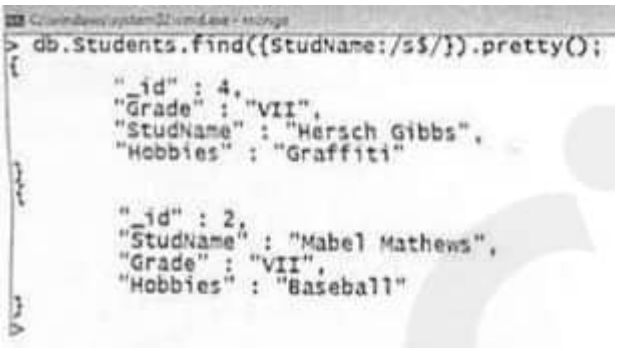
From Students

Where StudName like 'M%';

Objective: To find documents from the Students collection where the StudName ends in "s".

Act:

db.Students.find({StudName:/s$/}).pretty();




**RDBMS Equivalent:**

Select * From Students

Where StudName like '%s';

Objective: To find documents from the Students collection where the StudName has an "e" in any position.

Act:

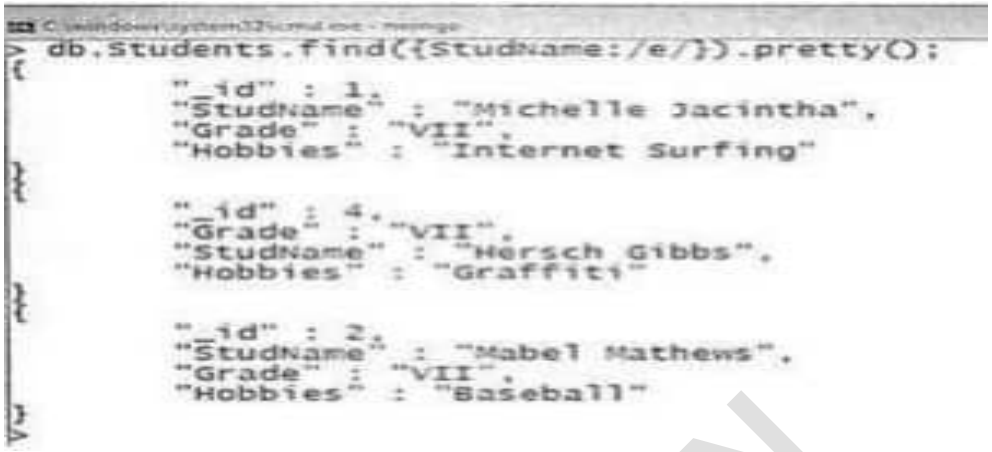
db.Students.find({StudName:/e/}).pretty();

OR

db.Students.find({StudName:/.*e.*/}).pretty();

OR

db.Students.find({StudName: {$regex:"e"}}).pretty();




**RDBMS Equivalent:**

Select * From Students

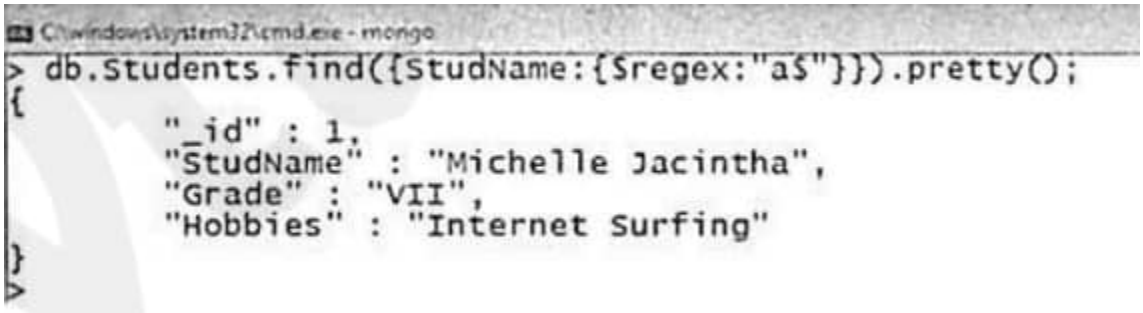
Where StudName like '%e%';

Objective: To find documents from the Students collection where the StudName ends in "a".

Act:

db.Students.find({StudName: {$regex:"a$"}}).pretty();

Outcome:




**RDBMS Equivalent:**

Select * From Students

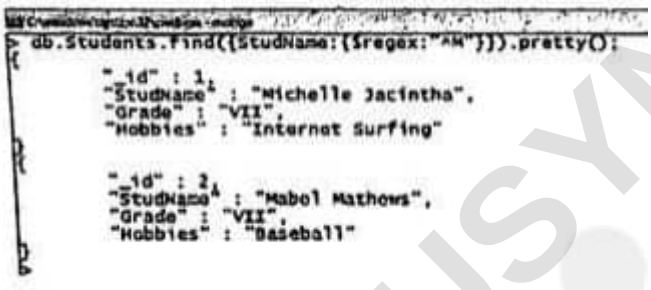
Where StudName like "%a";

Objective: To find documents from the Students collection where the StudName begins with "M".

Act:

db.Students.find({StudName:{$regex:"^M"}}).pretty();

**Outcome:**




**RDBMS Equivalent:**

Select * From Students

Where StudName like 'M%';

### 3.5.6 Dealing with NULL Values

**Objective:**

To add or manage a field (Location) with a NULL value in documents of the

Students collection.

• NULL indicates a missing or unknown value.

• This is useful when we don't know the value at the moment but may update it later.

**Step 1: Viewing Existing Documents**

To view specific documents before updating:

```
db.Students.find({$or: [{_id: 3}, {_id: 4}]})
```

```
db.Students.update({_id: 3}, {$set: {Location: null}});
```

```
db.Students.update({_id: 4}, {$set: {Location: null}});
```

**RDBMS Equivalent:**

```
UPDATE Students SET Location = NULL WHERE StudRollNo IN (3, 4);
```

**Step 3: Searching for NULL Values**

To find documents where Location is NULL or does not exist:

```
db.Students.find({Location: {$eq: null}});
```

**RDBMS Equivalent:**

```
SELECT * FROM Students WHERE Location IS NULL;
```

**Step 4: Removing Fields with NULL Values**

To remove the Location field where it's NULL:

```
db.Students.update({_id: 3}, {$unset: {Location: null}});
```

```
db.Students.update({_id: 4}, {$unset: {Location: null}});
```

**Step 5: Confirming the Change**

To verify that the fields have been removed:

```
db.Students.find()
```

• NULL values can be assigned using $set.

• NULL fields can be removed using $unset.

• Documents with NULL or missing fields can be queried with $eq: null

**3.5.7 Count, Limit, Sort, and Skip**

Objective: To find the number of documents in the Students collection.

Act:

```
db.Students.count()
```

Objective: To find the number of documents in the Students collection wherein the Grade is VII.

Act:

```
db.Students.count({Grade:"VII"});
```

**Step 2: Adding NULL Values**

To insert a NULL value in the "Location" field:

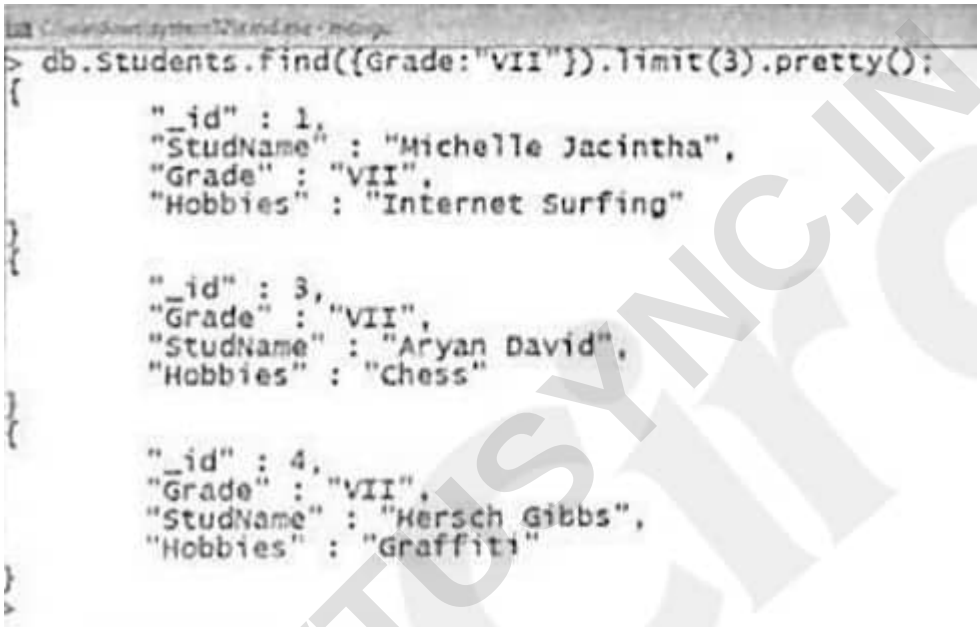
Objective: To retrieve the first 3 documents from the Students collection wherein the Grade is VII.

**Act:**

db.Students.find({Grade:"VII"}).

limit(3).pretty();

**Outcome:**




**RDBMS Equivalent:**

Select * From Students

Where Grade like "VII' and rownum <4;

Objective: To sort the documents from the Students collection in the ascending order of StudName.

Act:

db.Students.find().sort({StudName:1}).

pretty();

Outcome:

```
> db.Students.find().sort({StudName:1}).pretty();
{
        "_id" : 3,
        "Grade" : "VII",
        "StudName" : "Aryan David",
        "Hobbies" : "Chess"
}
{
        "_id" : 4,
        "Grade" : "VII",
        "StudName" : "Hersch Gibbs",
        "Hobbies" : "Graffiti"
}
{
        "_id" : 2,
        "StudName" : "Mabel Mathews",
        "Grade" : "VII",
        "Hobbies" : "Baseball"
}
{
        "_id" : 1,
        "StudName" : "Michelle Jacintha",
        "Grade" : "VII",
        "Hobbies" : "Internet Surfing"
}
{
        "_id" : ObjectId("5464849889ad1ab07d489b7f"),
        "StudName" : "Vamsi Bapat",
        "Grade" : "VI"
}
>
```

**RDBMS Equivalent:**

Select * From Students

Order by StudName asc;

Objective: To sort the documents from the Students collection in the descending order of StudName.

Act:

db.Students.find().sort((StudName:-1}).pretty();

**Outcome:**

```
> db.Students.find().sort({StudName:-1}).pretty();
{
        "_id" : ObjectId("5464849859ad1ab07d489b7f"),
        "StudName" : "Vansi Bapat",
        "Grade" : "VI"

        "_id" : 1,
        "StudName" : "Michelle Jacintha",
        "Grade" : "VII",
        "Hobbies" : "Internet Surfing"

        "_id" : 2,
        "StudName" : "Mabel Mathews",
        "Grade" : "VII",
        "Hobbies" : "Baseball"

        "_id" : 4,
        "Grade" : "VII",
        "StudName" : "Hersch Gibbs",
        "Hobbies" : "Graffiti"

        "_id" : 3,
        "Grade" : "VII",
        "StudName" : "Aryan David",
        "Hobbies" : "Chess"
```

**RDBMS Equivalent:**

Select * From Students Order by StudName desc;

Objective: To sort the documents from the Students collection first on Grade in ascending order and then on Hobbies in descending order.

Act:

db.Students.find().sort((Grade:1, Hobbies:-1)).pretty();